STANFORD UNIVERSITY
Department of Electrical Engineering

Lab 4: Sine shaper and system integration **Due:** Wednesday, 3 June 2026

Reading: *Function Generators*, commercial examples

Assignment: Complete your fabulous function generator by adding the sine shaper. Demonstrate that your generator provides sine, triangle and square waves from 2Hz up to 2MHz. Each team member must be present and ready to discuss in detail his or her individual contributions, as well as to answer questions.

Summary of specifications to be met:

Frequency range per band: There should continue to be no holes in coverage over the entire specified range.

Duty cycle error (square wave): When adjusted to zero at 10kHz, should continue not to exceed 2% at any other frequency, over the specified output voltage range of your buffer.

Triangle wave quality: The ratio of maximum to minimum amplitude should not exceed 1.1 over the entire frequency range. Any shift at peaks due to blow-by or any other mechanism should be below 2%. All of these are to be demonstrated with the buffer amplifier at a 2V nominal output amplitude into 50Ω.

Sine wave distortion: The formal specification is that the THD should not exceed 3% at any frequency, when measured at the output of your buffer at a 2V amplitude into 50Ω. Although you are not *required* to measure THD over the entire band, the waveform should always appear to be a "decent" sinewave to the eye (well, specifically, to the eye of the professor). The eye generally won't perceive a sinewave to be obviously nonsinusoidal until it has more than several percent distortion, so the eye spec is generous (and would lead to an uncompetitive commercial product). As a reference, note that a triangle wave has a THD of about 12%. In cases of persistent disagreement, a spectrum analyzer may be called into service as an impartial judge. The THD will be computed by considering only the first five harmonics.

Output amplitude: Up to 3.2V into 50Ω.

Supply voltage: ±12V

Allowed components: Your sine shaper should use parts drawn from the same collection of components you've been using all term.

Some possibilities: The Meyer-Sansen weakly-degenerated differential pair described in the notes is *strongly* recommended as a basis for your design. The diode-based circuits certainly can work, too, but the overwhelming experience of past students is that the diff. pair approach yields excellent results with much less effort than any other sine-shaping method described in the handout. You are, of course, free to select any method that you prefer. In the past, students have tried diode-LED and diode-resistor shapers, sometimes with transistor junctions as diodes. These efforts have shown that there are many possible ways to build a suitable shaper, although the amount of labor involved and suitability for volume production may differ.